Name ____________________ Math 1350, Quiz #1

1. (Section 1.2, Problem 17) Determine whether the three given points are the vertices of a right triangle

(9,6), (-5,4), (7,10)

17) (9,6), (-5,4), (7,10)

A B C

$\overline{AB} = \sqrt{(9 - {-5})^2 + (6-4)^2} = \sqrt{196 + 4} = \sqrt{200}$

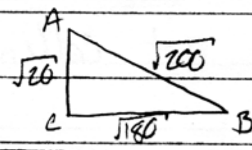
$\overline{BC} = \sqrt{(-5-7)^2 + (4-10)^2} = \sqrt{144 + 36} = \sqrt{180}$

$\overline{AC} = \sqrt{(9-7)^2 + (6-10)^2} = \sqrt{4 + 16} = \sqrt{20}$

$\sqrt{20}^2 + \sqrt{180}^2 = \sqrt{200}^2$

$20 + 180 = 200$

$\overline{AC}^2 + \overline{BC}^2 = \overline{AB}^2$



These points are the vertices of a right triangle.

2. (Section 1.3, Problem 9) Find the mid-point of the segment *AB*

A = (4,-1), *B* = (3,3)

9) A = (4, -1), B = (3, 3)

$x = \frac{x_1 + x_2}{2}$ $y = \frac{y_1 + y_2}{2}$

$= \frac{4+3}{2}$ $= \frac{-1+3}{2}$

$x = \frac{7}{2}$ $= \frac{2}{2}$

$y = 1$

$\boxed{\left(\frac{7}{2}, 1\right)}$

3. (Section 1.3, Problem 15) Find the point *P* between *A* and *B* such that the *AB* is divided into the given ratio.

$$A = (5, -3), \quad B = (-1, 6), \quad \overline{AP} / \overline{PB} = \tfrac{1}{2}$$

15.) $A = (5, -3)$, $B = (-1, 6)$, $\frac{\overline{AP}}{\overline{PB}} = \frac{1}{2}$

Let $\overline{AP} = 1/2$, $\overline{PB} = 1$. Then $\overline{AB} = 1/2 + 1 = \frac{3}{2}$. So $\frac{\overline{AP}}{\overline{AB}} = \frac{1/2}{3/2} = \frac{1}{3}$

$x = 5 + \frac{1}{3}(-1 - 5)$ $\qquad$ $y = -3 + \frac{1}{3}(6 - -3)$

$= 5 + \frac{1}{3}(-6)$ $\qquad$ $= -3 + \frac{1}{3}(9)$

$= 5 - 2$ $\qquad$ $= -3 + 3$

$x = 3$ $\qquad$ $y = 0$

$\boxed{P = (3, 0)}$